

The local Langlands categories

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Lecture 1. $G/\mathbb{Q}_p$ reductive, $l \neq p$.

$$\mathrm{Shv}\left([* / \underline{G}(\mathbb{Q}_p)], \overline{\mathbb{Q}_\ell}\right) \hookrightarrow \text{"\textit{Qcoh"}"}(\mathrm{Par}_G)$$

$$G \rightsquigarrow G' \text{ inner form } {}^L G \cong {}^L G' \rightsquigarrow \mathrm{Shv}\left([* / \underline{G}'(\mathbb{Q}_p)], \overline{\mathbb{Q}_\ell}\right)$$

need to interpolate $[* / \underline{G}'(\mathbb{Q}_p)]$ for inner forms of G

Kottwitz extended pure inner forms, forms arise from G -isocrystals (parametrized by $B(G)$)

$$\begin{array}{ccc} \text{Fargues-Scholze} & \xrightarrow{\quad} & \mathrm{Bun}_G \\ \text{Zhu} & \searrow & \mathrm{Isoc}_G \end{array}$$

1. Isocrystals with G -structures

$$1.1 \quad R \text{ perfect } \mathbb{F}_p\text{-alg.}, \quad W(R) \supset \sigma_R \quad \left(W(R) \rightsquigarrow W_{\mathbb{Q}_\ell}(R) \right)_{R \in \mathcal{D}}$$

Def. An $(F-)$ isocrystal over R is a pair (V, φ_V) where

V is a finite proj. $W(R)[\frac{1}{p}]$ -mod

$$\exists \varphi_V: \sigma_R^* V \xrightarrow{\sim} V \quad \text{iso.}$$

More: $V \rightarrow V'$ compatible w/ φ .

Denote^{by} $\mathrm{Isoc}(R)$ the cat. of isocrystals over R .

($\mathbb{Q}_p$ -linear, exact, $\otimes$ -cat, idempotent complete)

Ex 1 $R = \overline{\mathbb{F}_p}$

$$W(\overline{\mathbb{F}_p})[\frac{1}{p}]$$

$$\text{Isoc}(\overline{\mathbb{F}_p}) = \left\{ (V, \varphi_V) : V \text{ fin. dim. } \check{\mathcal{O}}_p\text{-vector space}, \varphi_V : \sigma^* V \xrightarrow{\sim} V \right\}$$

$$\Leftrightarrow \coprod_n GL_n(\check{\mathcal{O}}_p) / \sigma\text{-conj.} \quad \text{i.e. } b \sim b' \text{ iff } \exists g \text{ s.t. } gb\sigma(g)^{-1} = b'$$

(Deligne-Mumford classification) $\forall k = \bar{k} / \mathbb{F}_p$

$\text{Isoc}(k)$ is abelian, semisimple w/ simple obj. V_λ , $\lambda \in \mathbb{Q}$ (slope)

$$\lambda = \frac{s}{r}, \quad r > 0, \quad (r, s) = 1. \quad V_\lambda = \check{\mathcal{O}}_p^r, \quad \varphi = \begin{bmatrix} 0 & & & \\ & \ddots & & \\ & & 1 & \\ p^s & & & 0 \end{bmatrix}$$

Ex 2 A/R abelian scheme. $(V = H_{\text{cris}}^1(A/W(R))[\frac{1}{p}], \varphi_V) \in \text{Isoc}(R)$

Def. A G -isocrystal over R is a $\mathcal{O}_p$ -linear exact $\otimes$ -functor

$$\text{Rep}_{\mathcal{O}_p}(G) \rightarrow \text{Isoc}(R)$$

$\text{Isoc}_G(R) :=$ groupoid of G -isocrystals over R .

Def. $B(G) =$ set of isom. classes in $\text{Isoc}_G(\overline{\mathbb{F}_p})$ "Kottwitz set"

$$\xrightarrow{\sim} GL(\check{\mathcal{O}}_p) / \sigma\text{-conj.}$$

Steinberg th

$$\forall b \in B(G), \quad G_b : R/\mathcal{O}_p \hookrightarrow \left\{ g \in GL(\check{\mathcal{O}}_p \otimes_{\mathcal{O}_p} R) : gb\sigma(g)^{-1} = b \right\}$$

b is called basic if G_b inner form of G .

1.3 $\text{Isoc}(\overline{\mathbb{F}_p})$ is a $\mathcal{O}_p$ -linear (non-neutral) Tannakian category w/ a fibre functor over $\check{\mathcal{O}}_p$ (i.e. a $\mathcal{O}_p$ -linear rigid abelian $\otimes$ -cat w/ a $\mathcal{O}_p$ -linear paap?)

faithful exact $\otimes$ -functor in $\check{\mathcal{A}}_p$ -vect. spaces)

Deligne

$$\rightsquigarrow Kt_{\mathcal{A}_p} : R/\mathcal{A}_p \mapsto \text{Fun}_{\otimes}(\text{Isoc}(\bar{\mathbb{F}}_p), \text{VB}(R))$$

is an affine gerbe / $\mathcal{A}_p$

the "Kottwitz gerbe" for $\mathcal{A}_p$

Th (Iakovenko) $Kt_{\mathcal{A}_p}$ is banded by the profunctors:

$$ID := \varprojlim_{\substack{F(\mathcal{A}_p \text{ finite}) \\ \mathcal{Z} \mapsto \mathcal{Z}[F:\mathcal{A}_p]}} \mathcal{A}_m$$

(i.e. $Kt_{\mathcal{A}_p} \rightarrow \text{Spa } \mathcal{A}_p$ is a torsor under ID)

& its class in $H_{\text{ét}}^2(\mathcal{A}_p, ID)$ agrees w/ the image of the fundamental class

of local class field theory via

$$\begin{array}{ccc} \hat{\mathbb{Z}} & \xrightarrow{\sim} & H_{\text{ét}}^2(\mathcal{A}_p, \varprojlim_n \mu_n) \longrightarrow H_{\text{ét}}^2(\mathcal{A}_p, ID) \\ \downarrow \psi & \longmapsto & \\ 1 & & [Kt_{\mathcal{A}_p}] \end{array}$$

Lem. $B(\mathcal{A}) \simeq H_{\text{ét}}^1(Kt_{\mathcal{A}_p}, \mathcal{A})$

Pf. $H_{\text{ét}}^1(Kt_{\mathcal{A}_p}, \mathcal{A}) = \text{Fun}_{\otimes}(\text{Rep}_{\mathcal{A}_p}(\mathcal{A}), \text{VB}(Kt_{\mathcal{A}_p})) / \sim \quad (= B(\mathcal{A}))$

$$\begin{aligned} & \quad \quad \quad \text{is} \\ & \quad \quad \quad \text{VB}\left(\left[\frac{\text{Spa } \check{\mathcal{A}}_p}{\sigma\text{-}\mathbb{Z}}\right]\right) \\ & = H^1\left(\left[\frac{\text{Spa } \check{\mathcal{A}}_p}{\sigma\text{-}\mathbb{Z}}\right], \mathcal{A}\right) \simeq \mathcal{A}(\check{\mathcal{A}}_p) / \sigma\text{-conj.} \end{aligned}$$

Newton map: $k^t_{\mathcal{O}_p} \xrightarrow[\mathfrak{f}]{\text{IBID-torsion}} \text{Spa } \mathcal{O}_p$

Leray seq $0 \rightarrow H^1_{\text{ét}}(\mathcal{O}_p, G) \rightarrow H^1_{\text{ét}}(k^t_{\mathcal{O}_p}, G) \rightarrow H^0(\mathcal{O}_p, R^1 f_* G) \rightarrow$

$\xrightarrow{\text{is}} \underline{\text{Hom}}(1_D, G)/\text{Ad } G$

$\xrightarrow{\text{is}} \left(\text{Hom}_{\overline{\mathcal{O}_p}}(1_D, G) / \mathcal{U}(\overline{\mathcal{O}_p}) \right)^{\Gamma_{\mathcal{O}_p}}$

$\xrightarrow{\text{is}} \left(X^+_*(G)_{\mathcal{O}_p} \right)^{\Gamma_{\mathcal{O}_p}}$

$B(G) \xrightarrow{\nu_G}$

(Rem If $G = GL_n$, by Hilbert 90, ν_G is injective.)

Kottwitz map: consider the crossed module

(Fargues)

$$\begin{array}{ccc} [G_{\text{sc}} \rightarrow G] & & \\ \uparrow \text{ simply conn. } & \uparrow \text{ cover of } G_{\text{der}} & \\ [T_{\text{sc}} \rightarrow T] & & \end{array}$$

quasi-iso. —

$$G \curvearrowright G_{\text{der}} \leftarrow G_{\text{sc}}$$

LES $\hookrightarrow H^1(k^t_{\mathcal{O}_p}, G_{\text{sc}}) \rightarrow H^1(k^t_{\mathcal{O}_p}, G) \rightarrow H^1(k^t_{\mathcal{O}_p}, [G_{\text{sc}} \rightarrow G])$

$\uparrow \qquad \qquad \qquad \uparrow \qquad \qquad \qquad \uparrow$

$H^1(k^t_{\mathcal{O}_p}, T_{\text{sc}}) \rightarrow H^1(k^t_{\mathcal{O}_p}, T) \rightarrow H^1(k^t_{\mathcal{O}_p}, [T_{\text{sc}} \rightarrow T])$

$\parallel \qquad \qquad \qquad \parallel \qquad \qquad \qquad \parallel$

$B(T_{\text{sc}}) \simeq X_*(T_{\text{sc}})_{\mathcal{O}_p} \rightarrow B(T) \simeq X_*(T)_{\Gamma_{\mathcal{O}_p}} \rightarrow \pi_1(G)_{\Gamma_{\mathcal{O}_p}}$

$\rightarrow H^2(k^t_{\mathcal{O}_p}, T_{\text{sc}}) \stackrel{\text{Fargues}}{=} 0$

$$k_G: B(G) \rightarrow \pi_1(G)_{\Gamma_{\mathcal{O}_p}}$$

In (Kottwitz) $\textcircled{1} B(G) \xrightarrow{(\nu_G, k_G)} (X^+_*(G)_{\mathcal{O}_p})^{\Gamma_{\mathcal{O}_p}} \times \pi_1(G)_{\Gamma_{\mathcal{O}_p}}$ is injective

$$\textcircled{2} \quad B(\mathfrak{g})_{\text{basic}} = \{ b \in B(\mathfrak{g}) : v_{\mathfrak{g}}(b) \text{ central} \} \xrightarrow[\cong]{k_{\mathfrak{g}}} \pi_1(\mathfrak{g}) \cap \mathfrak{a}_p$$

Rem 1) $H^1(\mathfrak{a}_p, \mathfrak{g}) \cong \ker(v_{\mathfrak{g}}) \subset B(\mathfrak{g})_{\text{basic}}$

$\uparrow$
 $\mathfrak{a}_p$ pure inner forms

$\uparrow$
 ext'd pure inner forms

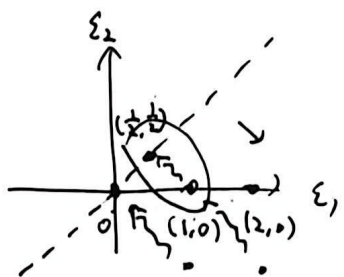
2) $k_{\mathfrak{g}}$ "first Chern class" of a $\mathfrak{g}$ -bundle on $k^+ \mathfrak{a}_p$

1.3 Def $b, b' \in B(\mathfrak{g})$, $b \leq b'$ if $k_{\mathfrak{g}}(b) = k_{\mathfrak{g}}(b')$, $v_{\mathfrak{g}}(b) \leq v_{\mathfrak{g}}(b')$

($v_{\mathfrak{g}}(b') - v_{\mathfrak{g}}(b)$ nonnegative
 sum of pos. coroots)

$\leadsto$ order top. on $B(\mathfrak{g})$: $b \in \overline{\{b'\}}$ iff $b \leq b'$.

Ex. 1 $\mathfrak{g} = \mathfrak{gl}_2$



$X_{\pi}^+(\mathfrak{g})$

Newton polygons

Ex. 2 $\mathfrak{g} = \mathfrak{sp}_4$, $\mu = (1, 1, 0, 0)$

$$B(\mathfrak{g}, \mu) = \{ b^{\text{ext}} \rightsquigarrow b^{\text{int}} \rightsquigarrow b^{\text{basic}} \}$$



Symmetric Newton polygons

Lecture 2. The algebraic local Langlands category (Zhu)

$\text{CALg}_{\mathbb{F}_p}^{\text{perf}}$: cat of perfect $\mathbb{F}_p$ -alg $\ni R$, $\mathbb{D}_R := \text{Spec } W(R)$, $\mathbb{D}_R^* = \text{Spec } W(R)[\frac{1}{p}] \ni \sigma_R$

PARAS

$G/\mathcal{O}_p$ reductive, $LG: R \mapsto G(W(R)[\frac{1}{p}])$

Def. $\mathcal{B}(G) := [LG / \text{Ad}_\sigma LG]_{\text{ét}}$ ("Isoc $_G$ " in Zhu's notation)

$$\begin{aligned} \text{Ad}_\sigma: LG \times LG &\rightarrow LG \\ (g, h) &\mapsto gh \sigma(g)^{-1} \end{aligned} \quad \sigma\text{-conj.}$$

$$\begin{array}{ccc} \Leftarrow \mathcal{B}(G) & \longrightarrow & \text{IBLG} \\ \downarrow & \nearrow & \downarrow \Delta \\ \text{IBLG} & \xrightarrow{\text{Id} \times \sigma} & \text{IBLG} \times \text{IBLG} \end{array}$$

Moduli interpretation: $R \mapsto \{(\mathcal{E}, \varphi): \mathcal{E} \text{ } G\text{-torsor on } \mathbb{D}_R^*, \varphi: \sigma_R^* \mathcal{E} \simeq \mathcal{E}\}$
+ étale loc. on R , $\mathcal{E}$ is trivial

Rem. In general for a $\overline{\mathbb{F}}^{\text{tr}}(Zariski)$ G -torsor on $\mathbb{D}_R^*$, we don't know if ét loc. on R

it can be trivialised, so (a priori) $\mathcal{B}(G) \neq \text{Isoc}_G := \text{Bun}_G(\mathbb{D}^*/\sigma\mathbb{Z})$

moduli of G -isocrystals

2.1 Lem: $\forall k_1 \subset k_2 / \mathbb{F}_p$ ext'n of alg closed fields, the nat'l map

$\mathcal{B}(G)(k_1) \rightarrow \mathcal{B}(G)(k_2)$ is an equiv. of groupoids, whose set of isom. classes

are both identified with $\mathcal{B}(G)$. [pf. ex]

Def. $b \in \mathcal{B}(G)$, $\mathcal{B}(G)_b \xrightarrow{i_b} \mathcal{B}(G)$ is the subfunctor sit.

$\text{Spec } R \rightarrow \mathcal{B}(G)$ factors through $\mathcal{B}(G)_b$ iff $\forall$ geom. pt $x \rightarrow \text{Spec } R$, the iso. class of $x \rightarrow \text{Spec } R \rightarrow \mathcal{B}(G)$ equals b . (Similarly one can define

$$\mathcal{B}(G)_{\leq b} \xrightarrow{i_{\leq b}} \mathcal{B}(G))$$

Th (Zhu, Th 3.31) Fix $k = \mathbb{F}_p$ & base change $\mathcal{B}(G)$ to k , then TFAT:

① $i_{\leq b}$ is perfectly finitely presented (pfp) closed immersion.

② $\overline{\mathcal{B}(G)_b} = \mathcal{B}(G)_{\leq b}$

* ③ $j_b : \mathcal{B}(G)_b \hookrightarrow \mathcal{B}(G)_{\leq b}$ is an affine open immersion.

④ $\mathcal{B}(G)_b \simeq \text{IB}_{\text{proét}} \underline{G_b(\mathcal{O}_p)}$, i.e. étale sheaf on $\text{CAlg}_{\mathbb{F}_p}^{\text{perf}}$

$R \mapsto$ the groupoid of proétale

$\underline{G_b(\mathcal{O}_p)}$ -torsors on $\text{Spec } R$.

Pr. $\mathcal{B}(G)_b \rightarrow \mathcal{B}(G)$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ \text{Isoc}_{G,b} & \longrightarrow & \text{Isoc}_G \end{array}$$

so suffices to prove for Isoc_G

① + ② & open part of ③, Grothendieck, Katz, Rapoport - Richartz

③ follows from purity of Newton stratification of F-isocrystals

(de Jong - Oort 00, ..., Hamacher - Kim 20)

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Th. A noeth local ring char p , $\dim A \geq 2$, $\xi \in \text{Isoc}(A)$ s.t.

$\text{NP}(\xi)$ constant on $\text{Spec } A \setminus \{m_A\} \Rightarrow \text{NP}(\xi)$ is constant.

Newton polygo.

$\Leftrightarrow \underline{A_{\text{is}}((\xi_b, \varphi_b))} = \underline{G_b(\mathcal{O}_p)}$

④ $(\xi, \varphi) \mapsto \underline{\text{Isom}}_{\text{proét}}((\xi, \varphi), (\xi_b, \varphi_b))$ where (ξ_b, φ_b) is the isotrivial

G -bundle on $[\mathbb{D}^*/\sigma^{\mathbb{Z}}]$ obtained via pullback along $[\mathbb{D}^*/\sigma^{\mathbb{Z}}] \rightarrow [\mathbb{D}_k^*/\sigma_k^{\mathbb{Z}}]$,

where $\text{Bun}_G([\mathbb{D}_k^*/\sigma_k^{\mathbb{Z}}]) \simeq \mathcal{B}(G)(k)$

proét loc. has a section due to Oort-Zink, Caraiani - Scholze, Hamacher - Kim

2.2 An ind- ptp proper chart

Def A morphism $X \xrightarrow{b} Y$ of ét stacks on $\text{CAlg}_{\mathbb{F}_p}^{\text{port}}$ is ind- ptp proper if

$\forall S \in \text{Sch}_{\mathbb{F}_p}^{\text{port}}$, $S \rightarrow Y$, $b_S: X_S \rightarrow S$ has a presentation $X_S = \varinjlim_i X_i \rightarrow S$
 $\text{AlgSP}_{\mathbb{F}_p}^{\text{port}} \quad \text{AlgSP}_{\mathbb{F}_p}^{\text{port}}$

wt $X_i \rightarrow X_j$ ptp closed immersion & $b_i: X_i \rightarrow S$ ptp proper.
 $\text{AlgSP}_{\mathbb{F}_p}^{\text{port}}$

Let $\mathcal{P}/\mathbb{Z}_p$ be a parahoric model of G

$$L^+ \mathcal{P} := R \mapsto \mathcal{P}(w(R))$$

$\omega_{\mathcal{P}} := LG / L^+ \mathcal{P}$. affine flag variety (ind $\text{proj.}/\mathbb{F}_p$)

$\text{Sh}^{\text{loc}}_{\mathcal{P}} := [LG / \text{Ad}_G L^+ \mathcal{P}] : R \mapsto$ groupoid of $\mathcal{P}$ -shukas over R

$$\text{i.e. } (\xi, \varphi: \sigma^* \xi|_{\text{ID}_R^*} \rightrightarrows \xi|_{\text{ID}_R^*})$$

$\uparrow$
 $\text{Bun}_{\mathcal{P}}(\text{ID}_R)$

Rem. ξ is automatically ét loc on R trivial since

$I \rightarrow w(R) \rightarrow R$, $(w(R), I)$ henselian pair & $\mathcal{P}$ -smooth

$$\text{Nt}_{\mathcal{P}} : \text{Sh}^{\text{loc}}_{\mathcal{P}} \rightarrow \mathcal{B}(G), \quad (\xi, \varphi) \mapsto (\xi|_{\text{ID}^x}, \varphi)$$

Prop. $\text{Nt}_{\mathcal{P}}$ is ind- ptp proper & surjective.

Pf. $\forall \text{Spec } R \rightarrow \mathcal{B}(G)$ con. to (ξ, φ) , $\exists R \rightarrow R'$ ét cover st. $\xi|_{\text{ID}_{R'}^*}$ trivial

$$\begin{array}{ccc} \omega_{\mathcal{P}, R'} & \longrightarrow & \text{Sh}^{\text{loc}}_{\mathcal{P}} \\ \downarrow & \searrow & \downarrow \\ \text{Spec } R' & \longrightarrow & \mathcal{B}(G) \end{array} \Rightarrow \text{claim.}$$

2.3. ℓ -adic sheaves

$$\Lambda = \mathbb{F}_\ell, \mathbb{Z}_\ell, \mathbb{Q}_\ell, \overline{\mathbb{Q}_\ell}$$

Basic goal: ① $\mathrm{Shv}(\mathrm{IB}_{\mathrm{pro\acute{e}t}} \underline{G(\mathbb{Q}_\ell)}, \Lambda) \simeq \underset{\text{it}}{\mathrm{D}(G(\mathbb{Q}_\ell), \Lambda)}$

define $\mathrm{Shv}(-, \Lambda)$ left completion at $\mathrm{D}(\mathrm{Rep}^{\mathrm{sm}}(G(\mathbb{Q}_\ell)))$

② descent + ℓ -functors

③ semi-orthogonal decomposition given by excision.

Naive approach: $\Lambda = \Lambda_0[\frac{1}{\ell}]$, Λ_0 ℓ -complete

$$\mathrm{D}(-, \Lambda_0) := \varprojlim_n \mathrm{D}(-, \Lambda_0/\ell^n)$$

$$\mathrm{D}(-, \Lambda) := \text{isogeny cat. of } \mathrm{D}(-, \Lambda_0)$$

Problem. eg. $M = \bigoplus_n \mathbb{Z}/\ell^n \xrightarrow{\otimes \mathbb{Q}_\ell} 0$

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$$\hat{M} \text{ has a free part } \xrightarrow{\otimes \mathbb{Q}_\ell} \neq 0$$

$\Lambda = \mathbb{Z}_\ell$ or $\mathbb{Q}_\ell$: $X \rightarrow \mathrm{Spa}(\overline{\mathbb{F}_\ell})$ ℓ -adic perfect scheme

$$\begin{array}{c} \downarrow \\ \varprojlim_{\alpha \in A} X_\alpha \end{array} \nearrow \quad X_\alpha \text{ PTP + affine transition maps}$$

$$\mathrm{Shv}(X_\alpha, \Lambda) := \mathrm{Ind} \mathrm{D}_c^b(X_\alpha, \Lambda)^{\mathrm{op}}$$

$$\mathrm{Shv}(X, \Lambda) := \mathrm{colim}_{\alpha \in A} \mathrm{Shv}(X_\alpha, \Lambda) \quad \text{w/ transition maps given by } !\text{-pullback.}$$

In general, right Kan extend to $X \in \mathrm{PreStk}_{\overline{\mathbb{F}_\ell}}$: $\mathrm{Shv}(X, \Lambda) = \lim_{S \rightarrow X} \mathrm{Shv}(S, \Lambda)$

Page 9 transition maps $= !\text{-pullback}$ $S \in (\mathrm{Sch}_{\ell\text{-adic}}^{\mathrm{pro\acute{e}t}})^{\mathrm{op}}$

Rem. 0) Depends on $X \rightarrow \mathrm{Spec} \overline{\mathbb{F}_p}$

1) if $X = \varinjlim X_\alpha$, $X_\alpha \rightarrow X_{\alpha'}$ representable ptp proper.

then $\mathrm{Shv}(X, \Lambda) \simeq \varinjlim \mathrm{Shv}(X_\alpha, \Lambda)$ via $*$ -pushforward.

2) has ind-ptp proper descent

3) eg. $X = \mathrm{Spec} \overline{\mathbb{F}_p}$,

$$\mathrm{Shv}(X, \Lambda) = \mathrm{Ind} \mathrm{Perf}(\Lambda) = D(\Lambda) \quad \swarrow \text{as a discrete ring}$$

eg. $X = \mathrm{IB}_{\mathrm{proet}} K$, K profinite group admitting a Λ -valued (Haar measure,

$\mathrm{Spec} \overline{\mathbb{F}_p} \simeq * \rightarrow X$ satisfies universal homological descent

Pt. use Bar-Beck-Lurie

$$\hookrightarrow \varinjlim (\dots K^n \rightrightarrows K^{n-1} \rightrightarrows \dots \rightrightarrows K \rightrightarrows *) \Rightarrow \mathrm{IB} K \neq X$$

Lecture 3. $G/\mathbb{Q}_p$

Rem G -torsor on $\mathrm{ID}_R = \mathrm{Spec} W(R)$ are ét loc. triv. on R

$$\mathrm{ID}_R^* = \mathrm{Spec} W(R)[\frac{1}{p}] \quad \text{not known}$$

$$G\text{-bundles} \left(\left[\mathrm{ID}_R^* / \mathcal{O}_R^\times \right] \right) = \mathrm{Isoc}_G(R)$$

$\xrightarrow{\quad}$

$\mathrm{Perf} \mathbb{F}_p$: cat of affinoid perfectoid spaces in char p

$\downarrow$
 $S = \mathrm{Spa}(R, R^+)$, $w \in R$ pseudo-uniformizer, $w(R^{(+)}) \subseteq \mathcal{O}_R$

$$\mathrm{FF}_S := \mathrm{Spa}(w(R^+)) \setminus V(p \cdot [w]) / \mathcal{O}_R^\times \rightarrow \mathrm{Spa} \mathbb{Q}_p \quad \text{adic space}$$

deg 1 Cartier divisors $\longleftrightarrow \mathcal{O}_R$ -orbits of units of S

$\text{Bun}_G : \text{Perf}_{\mathbb{F}_p} \rightarrow \text{Groupoids}, S \mapsto G\text{-bundles on } \text{FF}_S.$

(stack for the v -topology on $\text{Perf}_{\mathbb{F}_p}$)

3.1 For $S = \text{Spa}(R, R^+) / \overline{\mathbb{F}_p}$, have $\overline{\mathbb{F}_p} \rightarrow R^+$ induces

$$\text{FF}_S \xrightarrow{t_S} \left[\frac{\text{Spa } W(\overline{\mathbb{F}_p}) \setminus V(p)}{\sigma \mathbb{Z}} \right] = \left[\frac{\text{Spa } \mathbb{C}_p}{\sigma \mathbb{Z}} \right]$$

$$\rightsquigarrow \mathcal{B}(G)(\overline{\mathbb{F}_p}) \xrightarrow{t_S^*} \text{Bun}_G(S)$$

$$\ell \rightsquigarrow \overset{\vee}{E}_\ell \longmapsto \xi_\ell := t_S^* E_\ell$$

Th (Fargues-Anschütz) For $S = \text{Spa } C, C = \overline{C}$, the above gives a bijection

$$\mathcal{B}(G) \xrightarrow{\sim} \text{Bun}_G(C) / \simeq$$

⚠ Not an equiv. of cat since ξ_ℓ has more automorphisms!

$\rightsquigarrow \forall S \text{ (over } \mathbb{F}_p), \text{ given } \xi \text{ } G\text{-bundle } \text{FF}_S, \text{ get a function}$

$$|S| \rightarrow \mathcal{B}(G)$$

is

$$\left\{ \begin{array}{l} \text{Spa } C \rightarrow S \\ C = \overline{C} \end{array} \right\} / \simeq$$

$$\downarrow \\ S \mapsto \mathcal{B}[\xi_S]$$

FS : continuous if put opp. top. on $\mathcal{B}(G)$.

This induces $\text{Bun}_G \rightarrow \underline{\mathcal{B}(G)^{\text{op}}} - v\text{-sheaf attached to } \mathcal{B}(G)^{\text{op}}$

Def. $\text{Bun}_G^b := \text{Bun}_G \times_{\underline{B(G)^{\text{op}}}} \{b\}$, $\forall b \in B(G)$

Th (Fargues - Scholze, Viehweg),

① $|\text{Bun}_G| \simeq B(G)^{\text{op}}$

② $\text{Bun}_{G, \mathbb{F}_p}^b \simeq |\mathcal{B} \tilde{G}_b|$, where $\tilde{G}_b = \underline{A}^{+}(\varepsilon_b) \rightarrow \pi_0(\tilde{G}_b) \simeq \underline{G}_b(\mathbb{Q}_p)$

③ The semistable locus is $\text{Bun}_G^{\text{ss}} = \coprod_{b \in B(G)_{\text{basic}}} \left[* / \underline{G}_b(\mathbb{Q}_p) \right]$

Pr of ②: Need to show S strictly for. disc.

ξ on FF_S s.t. $b[\xi_S] = b$, $\forall s \in S$, $\xi \simeq \xi_b$.

Step 1. Reduce to the semistable case.

$H^1(\text{FF}_S, \mathcal{O}(\lambda)) = 0$ for S affinoid, for $\lambda > 0$.

Step 2 reduce to slope 0 using $\text{Bun}_G^b \simeq \text{Bun}_{G_b}^1$ for b basic.

Step 3 uses Kedlaya - Liu

$\left\{ \begin{array}{l} \text{semistable vector bundle} \\ \text{on } \text{FF}_S \text{ w/ ptwise slope } 0 \end{array} \right\} \simeq \left\{ \begin{array}{l} \text{prot } \underline{G}(\mathbb{Q}_p)\text{-loc} \\ \text{sys on } S \end{array} \right\}$

$\mathbb{L} \otimes_{\mathbb{Q}_p} \mathcal{O}_{\text{FF}_S} \hookrightarrow \mathbb{L}$

$\xi \longmapsto \text{BC}(\xi) := \bigoplus_S^1 \mapsto H^0(\text{FF}_S^1, \xi)$

So prot loc. triv, in part triv on S .

Rem / Digression : S perfectoid prot $\mathbb{Q}_p$ -loc. sys. on S & $\forall S^{\sharp}/\mathbb{Q}_p$ are the same, so VB on FF_S related to $\mathbb{Q}_p$ -loc. sys. on $S^{\sharp}$ (char 0)

can be used to study Gal rep.

eg. Th (Colmez - Fontaine) $K \supset \overset{\sim}{K_0} \supset \mathbb{Q}_p$ k
 $\downarrow$
 max unram. subfield

$$\text{Rep}_{\mathbb{Q}_p}^{\text{crys}}(\Gamma_K) \simeq \left\{ \underbrace{(V, \varphi)}_{\text{isocrystal over } k}, \underbrace{\text{Fil}^\bullet}_{\text{fil. on } V \otimes_{K_0} K} : \text{weakly admissible} \right\}$$

$\Downarrow$
 semistable for a suitable
 HN formalism

pt idea $\text{Spa } C \hookrightarrow \text{FF}_{C,b}$, $C = \widehat{\overline{K}}$

Both are identified w

$$\lim \left(\begin{array}{ccc} \varphi\text{-FilMod}_{K/K_0} & & \rightarrow (V, \varphi, \text{Fil}^\bullet) \\ \downarrow & & \downarrow \\ \text{Rep}_{\mathbb{Q}_p}(\Gamma_K) \longrightarrow \text{Bun}(\text{FF}_{C,b})^{\Gamma_K} & & \rightarrow \{ (V, \varphi), \mathbb{E} = \text{Fil}^\bullet(V \otimes_{K_0} B_{dR}) \} \\ \omega \longmapsto \omega \otimes_{\mathbb{Q}_p} \mathcal{O}_{\text{FF}_{C,b}} & & \uparrow \\ & & \hat{\mathcal{O}}_{\text{FF}, \infty} \text{-lattice} \end{array} \right)$$

2) Over $\mathbb{F}_p$, Bun_G^b is nontrivial gerbe

3.2 e.g. (Fargues) $G = \text{GL}_n$, $\text{Perf}_{\overline{\mathbb{F}_p}}$, $\Lambda / \mathbb{Z}_\ell$ -torsion

$$\text{Bun}_G = \text{Pic} = \coprod_{d \in \mathbb{Z}} \text{Pic}^d \simeq [x / \mathbb{A}_p^\times] \quad \mathcal{O}(d) = \{ (\check{\mathcal{O}}_p, p^{-d}) \}$$

$$\text{Div}^1: S \mapsto \left\{ \begin{array}{l} \deg 1 \text{ Cartier} \\ \text{div on } \text{FF}_S \end{array} \right\} \simeq \frac{\text{Spd } \check{\mathcal{O}}_p}{\mathbb{Z}}$$

$$\text{Div}^d \simeq (\text{Div}^1)^d / \mathbb{A}_p^d \quad \text{AJ}^d: \text{Div}^d \rightarrow \text{Pic}^d, D \mapsto \mathcal{O}(D)$$

$$\leadsto (AJ^d)^* : \text{Loc Sys}_{\bar{e}+}(\text{Pic}^d, \Lambda) \rightarrow \text{Loc Sys}(\text{Div}^d, \Lambda)$$

$$\text{When } d=1, \quad \text{Div}^1 \simeq \frac{\text{Spd } \check{\mathcal{O}}_P}{\sigma \mathbb{Z}} \simeq \left[\frac{\text{Spd } \mathcal{O}_P}{W_{\mathcal{O}_P}} \right]$$

⚠ $W_{\mathcal{O}_P} \simeq \text{Spd } \mathcal{O}_P$ not the usual action

$$\text{Spd } \check{\mathcal{O}}_P = \text{Spd } \bar{\mathbb{F}}_P \times_{\text{Spd } \bar{\mathbb{F}}_P} \text{Spd } \mathcal{O}_P, \quad \sigma = \text{id} \times \phi_{\text{Spd } \mathcal{O}_P}$$

$$\text{Rep}^{\text{sm}}(\mathcal{O}_P^{\times}, \Lambda) \simeq \text{Loc Sys}([* / \mathcal{O}_P^{\times}], \Lambda) \xrightarrow{(AJ^+)^*} \text{Loc Sys}(\text{Div}^1, \Lambda) \simeq \text{Rep}^{\text{cts}}(W_{\mathcal{O}_P}, \Lambda)$$

Th (Fargues) For $d \geq 3$, AJ^d is a proét loc. triv. fibration w/ simply conn'd fibres $\Rightarrow (AJ^1)^*$ is an equiv. on invertible obj.

$$\forall \mathbb{L} \text{ rk 1 on } \text{Div}^1, \quad S_d \mathbb{L} := (\pi_{d*} \mathbb{L}^{\boxtimes d})^{\oplus d} \quad (\text{Div}^1)^d \xrightarrow{\pi_d} \text{Div}^d$$

descends
 $\leadsto F_d$ on Pic^d if $d \geq 2$.

Show that $F_{d \geq 2} := \coprod_{d \geq 2} \text{Pic}^d$ extends uniquely to F on Pic $m^* F \simeq F \boxtimes F$

$$\leadsto \text{Aut}(\mathbb{L}) := F_1.$$

$$\begin{array}{ccc} \text{3.3 } \text{Hck}_G := \left\{ (D, \hat{\varepsilon}_1, \varepsilon_2, \varepsilon_1 \xrightarrow{\varphi} \varepsilon_2) \right\} & \xrightarrow{q} & \text{Hck}_G^{\text{loc}} = \left\{ (D, \hat{\varepsilon}_1, \hat{\varepsilon}_2, \varphi) \right\} \\ \downarrow h_1 & \swarrow h_2 & \downarrow \checkmark \\ \varepsilon_1 & \text{Div}^1 & \text{G-bundles on FF} \\ & \downarrow & \text{meromorphic} \\ & \text{Bun}_G & \text{(iso. outside } D) \\ & \text{Bun}_G \times \text{Div}^1 & \text{G-bundle on } \hat{\mathbb{F}}^D \end{array}$$

$$\text{have } S: \text{Rep}_\Lambda(\hat{a}) \rightarrow D(\text{Hck}_G^{\text{loc}}, \Lambda)$$

$$v \longmapsto S_v$$

$$T_V: D_{\text{ét}}(\text{Bun}_G, \Lambda) \longrightarrow D_{\text{ét}}(\text{Bun}_G, \Lambda)^{B_W \rtimes \text{Ap}}$$

(Λ torsion) $F \longmapsto h_{2!}(h_1^* F \otimes q^* S_V)$

Rem. For $\Lambda = \mathbb{Z}_\ell, \mathbb{Q}_\ell, \dots$ X Artin

$$D_{\text{lis}}(X, \Lambda) \subset D_{\square}(X, \Lambda)$$

stable co-cat gen. under all colimits by $t_{\square} \Lambda$ for $t: Y \rightarrow X$ sep, representable,
 ℓ -coh smooth

3.4. Conj. (Fargues, rough version) $\Lambda = \overline{\mathbb{Q}_\ell}$, G quasi-split.

$\exists$ functor of groupoid

$$\text{Discrete } L\text{-par.} \longrightarrow \text{Perverse Weil sheaves on } \text{Bun}_G$$

$$\varphi \longmapsto F_\varphi \quad (\text{cf. BM-K})$$

s.t. ① $i_b^* F_\varphi$, b basic gives LLC for G_b

② If φ cuspidal, $i_b^* F_\varphi = 0$ unless b basic (known by FS)

$$\textcircled{3} T_V(F_\varphi) \simeq F_\varphi \boxtimes \underline{r_V \circ \varphi} \quad (\text{eigensheaf})$$

④ Compatibility w/ global Langlands (coh. of Shimura varieties)
 via Hodge-Tate period map. (cf. Yang, Hamann's talk)